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Climate Change Impact Analyzer

The Climate Change Impact Analyzer is a Python-based tool developed to examine historical climate data, providing insights into temperature trends, regional patterns, and anomalies. Designed for educational purposes, it integrates real-world data from the Meteostat API and employs three algorithms: linear regression, K-means clustering, and statistical anomaly detection. This report describes these algorithms, outlines their processes, explains the reason for their selection, and presents findings demonstrating their efficiency and benefits.

Algorithms

The project's algorithms, written in `algorithms.py`, form the analytical core. Linear regression, implemented via Scikit-learn's `LinearRegression` class in the `TemperaturePredictor` module, models the relationship between temporal features (year and month) and normalized temperature. It fits a linear equation to minimize squared errors, predicting future temperature trends. The design takes a linear progression, suitable for capturing gradual climate shifts.

K-means clustering, defined in the `custom_clustering` function, groups data into three clusters using Scikit-learn's `KMeans`. Operating on annual temperature mean and standard deviation, it minimizes intra-cluster variance through iterative point updates. This method identifies distinct climate patterns, such as stable or variable temperature regimes.

Statistical anomaly detection, implemented in `detect_anomalies`, flags outliers exceeding three standard deviations from the mean. This threshold-based approach targets extreme temperature events, such as heatwaves or data errors, without requiring complex model training.

Processes

The workflow, managed by `main.py` and `cli.py`, follows a structured pipeline: data acquisition, preprocessing, analysis, and visualization. The `DataProcessor` class in `data_processor.py` retrieves data, either from a CSV file (e.g., `climate_data_api.csv`) or the Meteostat API. The `API` method, `download_data_from_api`, fetches daily temperature records for user-specified locations and periods (having a default from 2000–2020), saving them with columns for time, year, month, and temperature.

Preprocessing, handled by `clean_data`, removes missing values and normalizes temperatures using z-score standardization. For prediction, `get_features_and_target` extracts year and month as features and normalized temperature as the target.

Clustering uses `get_features_for_clustering` to compute annual mean and standard deviation. Analysis options, which are prediction, clustering, or anomaly detection, are selected via an interactive menu or through command-line arguments.

Visualization, implemented in `visualizer.py`, uses Plotly to generate interactive HTML graphs. These include temperature trend plots comparing actual and predicted values, scatter plots labeling clusters (e.g., Stable Cold Climate), and anomaly plots marking outliers over time. Outputs are saved in the `htmls/` directory and opened automatically in a web browser.

Algorithm Selection

The algorithms were chosen for their simplicity, interpretability, and suitability for a class project. Linear regression was chosen for predicting temperatures because it effectively captures long-term trends. Its simple coefficients clearly show how temporal factors influence the results, making it particularly suitable for educational settings where understanding is more important than complexity. More sophisticated models, like neural networks, were not utilized because of their higher computational requirements and the linear nature of the dataset.

K-means clustering was adopted to uncover patterns in annual temperature data without requiring labeled inputs. The choice of three clusters balances granularity and clarity, enabling distinctions between stable and variable climates. Alternatives like

hierarchical clustering were less convenient given the dataset's size and the project's scope.

Statistical anomaly detection was chosen for its minimal computational requirements and effectiveness in identifying extreme events. By using a fixed threshold, it avoids the tuning complexity of machine learning-based methods, aligning with the project's goal of accessible climate analysis.

Findings on Efficiency and Benefits

Analysis of the dataset (`climate_data_api.csv`), spanning 2000–2020 with 7,670 daily temperature records, demonstrates the algorithms' efficiency and benefits. Linear regression, training in under a second, models annual mean temperatures (z-scores from -1.12 to 1.45), revealing a warming trend of approximately 0.05 z-scores per year (`interactive_temperature_trend.html`). The tight alignment of predicted and actual values in the trend plot suggests reliable forecasting, with low computational cost enabling scalability for larger datasets. The interactive visualization enhances accessibility, allowing users to explore trends intuitively.

K-means clustering processes 21 annual data points (2000–2020) in milliseconds, yielding three distinct groups (`interactive_clusters.html`): Stable Warm Climate (5 years, z-scores ~ 1.23 – 1.45 , standard deviation ~ 0.12 – 0.15 , e.g., 2019–2020), Stable Cold Climate (15 years, z-scores ~ -1.12 – 0.98 , standard deviation ~ 0.10 – 0.18), and Moderately Variable Climate (1 year, z-score ~ 0.05 , standard deviation ~ 0.95). This segmentation efficiently captures climate shifts, using minimal features to ensure robustness. The scatter plot's clear cluster separation aids interpretation, supporting educational goals.

Anomaly detection scans the dataset in milliseconds, identifying 382 outliers (z-scores ~ 3.0 – 4.5 , $\sim 5\%$ of days) exceeding three standard deviations (`interactive_anomalies_normalized.html`). Anomalies, such as January 21, 2000 (z-score ~ 3.07), cluster in winter months (e.g., January–March 2000–2001, February–March 2010), highlighting extreme events like heatwaves or cold snaps. The plot's red markers ensure clarity, minimizing false positives through normalization. Its speed suits real-time applications.

The algorithms' efficiency processes thousands of records rapidly, it supports their use in resource constrained environments. Benefits include linear regression's reliable trend prediction for climate planning, K-means' pattern detection for regional insights, and anomaly detection's ability to flag critical events for investigation. Interactive HTML outputs empower users, making the tool valuable for education and preliminary climate analysis.

The Climate Change Impact Analyzer shows how linear regression, K-means clustering, and statistical anomaly detection can study temperature data from 2000 to 2020. The tool's steps, starting from grabbing data, cleaning it, analyzing it, and ending by making graphs is a process that works fast and clearly. These algorithms were picked for being simple and good for learning, revealing a warming trend, three climate patterns, and 382 odd days, mostly in winter. The quick processing and easy-to-use HTML plots make it great for class projects and understanding climate shifts. This tool proves basic methods can help explore big climate questions in an efficient and simple manner.